FNCE30007 Tutorial 10 – Interest Rate Contracts and Swaps

Question 1
Suppose that zero interest rates with continuous compounding are as follows:

Maturity (years)	Rate (% per annum)
1	2.0
2	3.0
3	3.7
4	4.2
5	4.5

Calculate forward interest rates for the second, third, fourth, and fifth years.

Question 2
Explain the difference between the credit risk and market risk in a financial contract.

Question 3
Companies X and Y have been offered the following rates per annum on a $5 million 10-year investment:

	Fixed Rate	Floating Rate
Company X	8.0%	SOFR
Company Y	8.8%	SOFR

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.2% per annum and will appear equally attractive to X and Y.

Question 4.
A financial institution has entered into an interest rate swap with company X. Under the terms of the swap, it receives 4% per annum and pays six-month SOFR on a principal of $10 million for five years. Payments are made every six months. Suppose that company X defaults on the sixth payment date (end of year 3) when the six-month forward SOFR rates for all maturities are 2% per annum. What is the loss to the financial institution? Assume that six-month SOFR was 3% per annum halfway through year 3 and that at the time of default all zero rates are 1.8% per annum. Zero rates are expressed with continuous compounding; other rates are expressed with semiannual compounding.

Question 5.
Six-month SOFR is 5%. SOFR forward rates for the 6- to 12-month period and for the 12- to 18-month period are both 5.5%. Fixed rates for 2- and 3-year semiannual pay swaps are 5.4% and 5.6%, respectively. Estimate the SOFR forward rates for maturities of 18 months to 2 years, 2 to 2.5 years, and 2.5 to 3 years. Assume that the 2.5-year fixed swap rate is the average of the 2- and 3-year fixed swap rates.

Assume the continuously compounded zero rates are as follows:

Maturity	Zero Rate (cont. comp.)
0.5 year	4.20%
1.0 year	4.35%
1.5 years	4.50%
2.0 years	4.65%
2.5 years	4.80%
3.0 years	4.95%

All zero rates are expressed with continuous compounding; all other rates are expressed with semiannual compounding.